

kea: A Python package for consistent power spectral analysis of data on arbitrary dimensions

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Summary

Power spectral density techniques are widely used across disciplines but often differ in normalization and data handling, with most fields relying on a limited set of established methods. Historically, researchers have been required to implement their own routines or personally request code from others. This increases barriers to entry for performing analysis and requires a deep understanding of specific Fourier conventions to avoid errors. Without a standardized framework, order-unity normalization difference often creep in, which can lead to physical inferences that are mathematically inconsistent or physically erroneous. To address this, we present a computational package that provides consistently normalized estimators that are adaptable to multiple data types – including intracluster medium surface brightness fluctuations, in-situ solar wind time-series, and simulation cubes.

Statement of Need

Autocorrelation functions (ACF), power spectral densities (PSD), and structure functions (SF) serve as the primary mathematical framework for quantifying the statistical properties of stochastic phenomena across varying spatial and temporal scales. In the context of fluid dynamics and astrophysics, these tools allow researchers to decompose complex, multiscale signals into their constituent parts. Of particular interest, especially within the study of (magneto)-hydrodynamic turbulence, are the characteristic scales (such as the integral scale L where energy is injected, and the dissipation scale η), the power-law indices that define the energy cascade (e.g., the Kolmogorov $-5/3$), and their respective amplitudes which dictate the total turbulent energy budget.

These phenomena are rarely captured in the full continuum of values in the available space in which measurements occur. In other words, in a lot of practical cases, observations are restricted to only 1D slices, 2D projections, or 2D slices. For example, in the case of the solar wind, in-situ measurements are performed by sensors moving in relation to the plasma rest frame. These sensors generate a time-series of data: such as magnetic field (vector field), or density (scalar field) measurements. As another pertinent example, surface brightness observations of the intracluster medium (ICM) observe 2D (emission weighted) projections of scalar fields, or projections of line-of-sight components of a vector field. Often these observations are calibrated or compared against more accessible regimes, or high-fidelity simulations.

Beyond the choice of statistical tool, the reliability of the output is heavily dependent on the nuances of spectral estimation. There exists a vast library of signal processing techniques designed to mitigate the distortions, aliasing, and biases inherent in finite sampling ([Maciejewski](#)

et al., 2009; Sefusatti et al., 2016; Stoica & Moses, 2005). However, the implementation of these techniques introduces its own set of variables: how the data are windowed to prevent spectral leakage, how the estimates are binned in k -space, and how the resulting power is normalized (e.g., ensuring Parseval's theorem is satisfied). For example, there are a total of six different R packages that provide spectrum estimations with varying normalization options (Barbour & Parker, 2022).

If these methodological choices are not standardized, they can induce spurious physical behavior. For example, improper binning can artificially flatten a spectral slope. This could become a significant issue when comparing across different studies or when cross-correlating observational data with numerical models. Without a rigorous, consistent approach to these estimations, the resulting physical interpretations (such as the injection or dissipation scales) may reflect the limitations of the signal processing rather than the underlying physics.

We thereby introduce Kea (Analysis Engine MarK, backwards), a Python package that implements several different dimensionally-agnostic scale-dependent statistical estimation techniques for use in turbulence analysis. Kea is available on GitHub via <https://github.com/mab68/Kea>.

Power-law correlations are a key feature of turbulence (particularly with high Reynolds numbers flows). The scale-to-scale energy transfer associated with the cascade of energy from large scales to smaller scales leads to a power-law power spectrum which is indicative of fractal-like structure where small-scale features are statistically similar to large-scale ones.

Observing the complete set of information provided by self-similar physical phenomena can be practically challenging. In particular, we highlight that the majority of astronomy and astrophysical observations are driven by the detection and manipulation of photons. The properties of these photons are determined by the conditions of the source (temperature, density, velocity, etc.) of the emitting material and its subsequent interaction with any intervening material. Determining the conditions of the source from the photons alone can be a serious challenge. Noise and resolution constraints may also limit the accuracy of the inference and increase complexity. To understand biases and systematics or comparison of theory to observation, where data is lacking, it can be necessary to compute expected observational properties from controlled surrogate models (Haworth et al., 2018; Simionescu et al., 2019).

State of the Field

While the mathematical definitions of autocorrelation functions, power spectral densities and structure functions are well-established, their application to “real-world” datasets show significant variability in subtle ways. For example, normalizations of the difference-of-Gaussian method differ (Arévalo et al., 2012; Churazov et al., 2012; Zhou et al., 2022).

Currently available software for ICM turbulence analysis are: turbustat (Koch et al., 2019), and PITSZI (Adam et al., 2025). Both of these solutions require 2D data, thus are not available for applications to e.g., solar wind turbulence analysis, or simulation cubes. Additionally, they both use a nested sequence of object-oriented class based design with built-in handling with assumptions for their respective fields: the interstellar medium for turbustat, and the intracluster medium for PITSZI. Hence, modifying for individual needs is challenging. SF calculation is available using fastSF (Sadhukhan et al., 2021), however, only for uniform 2D and 3D datasets. In other words, fastSF is not applicable to gapped datasets. SF calculation is also available in Python with fluidsfs (Wagner et al., 2025).

Software Design

Kea is designed with two main goals: provide standardized tools for robust, multidimensional spectral estimation, and to generate controlled synthetic fields for validating these analytical

88 methods. To efficiently support these objectives, Kea is written as a Python package that
89 operates directly on NumPy arrays, ensuring it is easily integrable across varying fields of research.

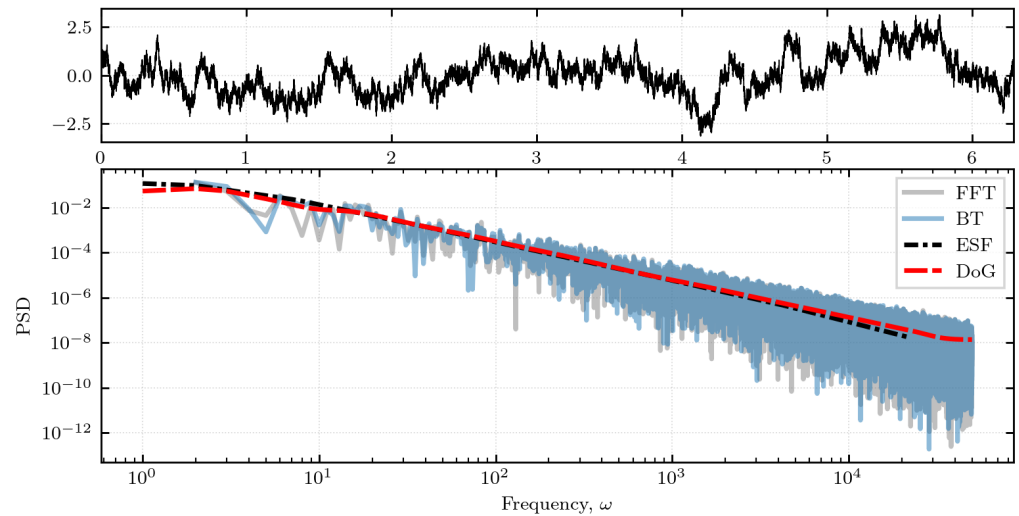


Figure 1: Spectral estimation capabilities of Kea. The top panel displays a 1D synthetic timeseries generated by the software, while the bottom panel illustrates the resulting spectral estimates computed using Kea's implementations of the periodogram (FFT), Blackman-Tukey (BT), equivalent spectrum (ESF), and difference-of-Gaussian (DoG) methods.

90 Kea provides discrete implementations of several key spectral estimation techniques. An
91 application of these methods to a synthetic timeseries is demonstrated in Figure 1. Since
92 the autocorrelation function (ACF), structure function (SF), and the FFT and BT spectral
93 estimates yield D -dimensional functions, it is often necessary to bin them to reduce their
94 dimensionality. To address this, Kea provides a general binning routine applicable to both the
95 spectral estimates and the lag-functions.

96 Furthermore, Kea provides dedicated utilities for generating synthetic spatial fields.

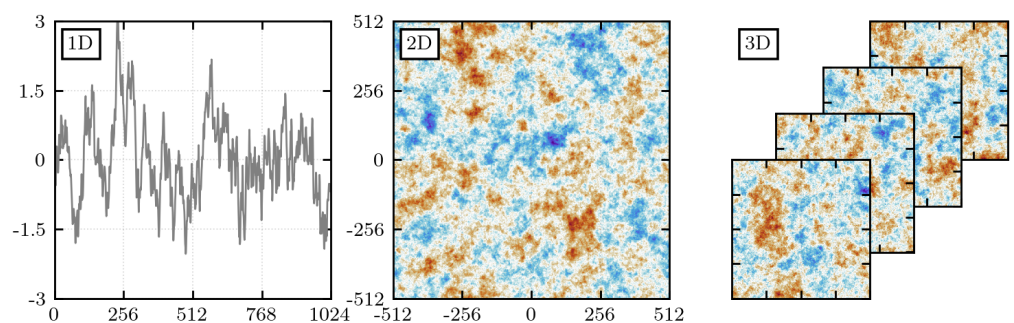


Figure 2: Synthetic monofractal test fields generated by Kea, demonstrating the package's support for 1D (left), 2D (middle), and 3D (right, shown as a 2D slices) spatial data structures. These generation tools allow users to easily create controlled datasets for testing and benchmarking.

97 Synthetic fields have been used extensively to validate methods and measure turbulence in
98 the interstellar medium (Brunt & Heyer, 2002; Esquivel et al., 2003; Miville-Deschênes et
99 al., 2003; Ossenkopf et al., 2006) and similarly for the ICM (Vogt & EnBlin, 2005; XRISM
100 Collaboration et al., 2025; ZuHone et al., 2016). Kea provides methods to produce (monofractal

and multifractal) synthetic fields with correlations parameterized by power spectral density. These fields are synthesized in Fourier-space and Fourier-transformed to provide real-space fluctuation fields (Barnsley et al., 1988; Lakhal et al., 2025).

For examples, shown in Figure 2 are one, two, and three-dimensional monofractal fields. Thus, the below PSD estimators can be tested for validity with known/expected forms, and their performance tested under constraints of noise and missing data.

Research Impact Statement

AI Usage Disclosure

The paper was initially drafted in full by the authors. Google Gemini 3.1 was used after the drafting of this manuscript to assist with linguistic polishing to improve manuscript clarity. Google Gemini 3.1 was also used to suggest improvements to the code. All AI-assisted text and code were thoroughly reviewed, tested, and verified by the authors to ensure correctness. The authors assume full responsibility for the final content and scientific accuracy of the manuscript and code.

Future Work

It is intended for future development of kea to implement a general interpretation of spectral estimation using arbitrary filters along with automatic implementation of debiasing using the non-parametric local power-law approximation that is currently only available for the ESF method.

Another area of potential improvement will be the implementation of synthetic fields with anisotropy. This would be useful to test the statistics of projected fluctuations in more complex scenarios.

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